QISNet
ICCCN 2027
WORKSHOP

1ST INTERNATIONAL WORKSHOP

Quantum Intelligence, Security, and Networking (QISNet 2027)

In conjunction with the 36th International Conference on Computer Communications and Networks (ICCCN 2027) · Rome, Italy · Workshop date: TBA

CALL FOR PAPERS

Scope

The 1st International Workshop on Quantum Intelligence, Security, and Networking (QISNet 2027) aims to provide an interdisciplinary forum for researchers and practitioners exploring how quantum science and technology can transform future communication, networking, distributed computing, artificial intelligence, and cybersecurity.

QISNet 2027 focuses on the convergence of quantum computing, quantum artificial intelligence, quantum communications, and quantum security with next-generation computer networks and distributed systems. Quantum computing and quantum machine learning may enable new approaches to network optimization, resource allocation, traffic analysis, distributed intelligence, and complex decision-making. Quantum communication technologies—including quantum key distribution, entanglement distribution, quantum repeaters, and emerging quantum-network architectures—introduce fundamentally new models for secure and distributed communication. At the same time, advances in quantum computing motivate a transition toward post-quantum cryptography, crypto-agility, and quantum-resilient network infrastructures.

The workshop places a strong emphasis on communications and networking, including architectures, protocols, algorithms, security mechanisms, intelligent resource-management methods, experimental systems, testbeds, and applications that connect quantum technologies with computer communications and networks. The workshop welcomes theoretical, experimental, and systems-oriented work, including hybrid quantum-classical approaches.

Topics of Interest

Topics include, but are not limited to:

Quantum Computing and Quantum AI for Networks

- Quantum computing for communication and network optimization
- Quantum algorithms for routing, scheduling, resource allocation, and network design
- Quantum machine learning and quantum AI for networks
- Hybrid quantum-classical AI for networked and distributed systems
- Quantum-enhanced network traffic analysis, prediction, and anomaly detection
- Quantum-enabled optimization for edge, cloud, and distributed computing

Quantum Communication Networks and the Quantum Internet

- Quantum communication networks and architectures
- Quantum internet architectures, protocols, and services
- Quantum key distribution networks and secure key management
- Entanglement generation, distribution, routing, and resource management
- Quantum repeaters, quantum memories, and network control
- Quantum network simulation, modeling, emulation, and testbeds
- Classical-quantum network integration and control planes
- Quantum networking for data centers, edge systems, IoT, and cyber-physical systems

Quantum-Safe Security and Post-Quantum Cryptography

- Post-quantum cryptography for communication and network systems
- Crypto-agility and migration to quantum-safe network infrastructures
- Quantum-resilient authentication, identity, and trust management
- Secure protocol design for quantum and post-quantum networks

- Quantum threat modeling, risk assessment, and network security
- AI-assisted security for quantum and post-quantum systems

Quantum Sensing, Distributed Intelligence, and Systems

- Quantum sensing and networked quantum sensors
- Quantum-enhanced distributed computing and federated or collaborative intelligence
- Resource management and performance evaluation in quantum-enabled networks
- Reliability, scalability, interoperability, and standardization of quantum networks
- Experimental platforms, prototypes, applications, and use cases for quantum-enabled networks

Submission Guidelines

- Authors are invited to submit manuscripts reporting original unpublished research and recent developments related to the workshop topics.
- Submissions should include an abstract, keywords, and the e-mail address of the corresponding author.
- Papers are limited to **6 pages** in standard IEEE camera-ready format (double-column, 10-pt font).
- All submitted papers will be peer reviewed. Accepted papers must satisfy the final ICCCN 2027 registration, presentation, formatting, and publication requirements.
- Submission of a paper is regarded as an undertaking that, should the paper be accepted, at least one author will register for and attend the workshop to present the work.
- Accepted and presented workshop papers are expected to be included in the ICCCN conference proceedings and submitted for publication in IEEE Xplore, subject to the policies and technical co-sponsorship arrangements announced by ICCCN 2027.
- The submission link and detailed ICCCN 2027 submission instructions will be posted on the workshop website when available.

Important Dates

Paper submission deadline	TBA
Acceptance notification	TBA
Camera-ready due	TBA
Workshop date	TBA

All dates will be aligned with ICCCN 2027.

Workshop Chairs

Shuhui Yang, Purdue University Northwest, USA — shuhuiyang@pnw.edu
Yunhe Feng, University of North Texas, USA — Yunhe.Feng@unt.edu

Technical Program Committee (Tentative)

Celso Jorge Villas-Boas (Federal University of São Carlos); Shengwang Du (Purdue University); Lei Fan (University of Houston); Bo Fang (University of Texas at Arlington); Qiang Guan (Kent State University); Shafkat Islam (Purdue University Northwest); Kishor Kapale (Western Illinois University); Tae-Hoon Kim (Purdue University Northwest); David Q. Liu (Purdue University); Antonio Padua Paes (D-Wave); Runzhou Tao (University of Maryland); Yu Wang (Temple University); Shawn He Yuxun (Purdue University Northwest).